

Data Card

Latent Probing for Mental Health Sentiment Classification

Input vs. Output

Input: Short-to-medium length user-generated text posts (originally sourced from Reddit and Twitter/X), written in English.

Output: A predicted mental health sentiment label (Depression, Anxiety, Stress, or Normal), along with a latent probe signal used to interpret model confidence and reasoning, rather than a bare classification alone.

Classification Problem

Task: Multi-class text classification.

Classes: Depression, Anxiety, Stress, and Normal. These are a subset of the categories available in the source dataset, selected because they represent the most common, non-crisis-level distress signals relevant to early identification. The project intentionally excludes higher-risk categories such as "Suicidal" in order to keep scope focused on early, non-crisis support instead of acute crisis intervention.

Labeling

Source of labels: Original labels come from the "Sentiment Analysis for Mental Health" dataset (Kaggle), an aggregation of public Reddit and Twitter/X posts pre-labeled by the dataset's original curators.

Quality and consistency check: The team will draw a random sample of approximately 50 to 100 posts and independently re-label them using a rubric based on DSM-5 symptom descriptors (see table below). Inter-rater agreement between team members and the original labels will be calculated, and any disagreements will be discussed and documented as part of the project's quality process.

Label	Key indicators
Depression	Persistent sadness or emptiness, loss of interest, worthlessness, fatigue, hopelessness
Anxiety	Excessive worry, fear of losing control, restlessness, physical tension, dread about the future
Stress	Feeling overwhelmed, situational pressure (work, school, relationships), typically short term
Normal	None of the above patterns present

Data Creation

The dataset is not newly collected by the team; it is an existing, publicly available, aggregated dataset combining posts from Reddit and Twitter/X. No synthetic data generation is used.

Web Collection Ethics

The team is not scraping any new data from the web; only the existing, pre-published Kaggle dataset is used (secondary use of already-public data).

The original dataset creators sourced posts from public social media platforms (Reddit, Twitter/X, and Facebook) using platform APIs and web scraping tools, then removed direct identifiers (e.g., usernames) prior to publication.

Important limitation: the individuals whose posts make up this dataset were not asked for consent specifically to have their text used in mental health classification research. Their posts were public at the time of collection under each platform's terms of service, but that is not the same as informed consent for this specific research use. This is a known and common limitation across social media derived mental health datasets. The team mitigates this by using only anonymized, already published data, avoiding any attempt at re-identification, not publishing raw post excerpts that could expose a person, and restricting use of the dataset to educational and research purposes only, not commercial or public facing deployment.

Bias and Representation

Likely underrepresented groups: non-English speakers, older adults (Reddit and Twitter/X skew younger), people without internet or social media access, and individuals who never post publicly about mental health due to stigma, meaning silent sufferers may be systematically excluded from the very dataset meant to help detect them.

Mitigation: the team will document these known gaps transparently in the final report, avoid overstating generalizability of results, and recommend future work to expand coverage to underrepresented languages and offline populations.

Values and Community Alignment

The dataset selection and rubric-based validation process directly reflect the values named in the Challenge 1 problem statement: privacy (no re-identification, secondary use only), safety (human-in-the-loop validation, no autonomous diagnosis), fairness (explicit documentation of bias and demographic gaps), and transparency (interpretable latent probes, documented labeling process).

Community benefit: the tool is designed to support, not replace, early identification, flagging patterns for human review rather than making autonomous decisions. Accountability is maintained by planning to consult a psychology or counseling professional at the team's institution to review the labeling rubric before any real-world use is considered.